# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-LAMBERT-EN · v2.2 · 2026-11-02 |
| Package | star-lambert |
| Version | 0.1.3 |
| Capability | star.orbit.lambert |
| Maturity | Beta (VERIFIED) |
| Python | ⩾3.9 |
| Source commit | dc09f61 |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_lambert |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_lambert-0.1.3-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_lambert-0.1.3-py3-none-any.whl | eb900215a0a6fe0470efc8dcb76dcc5b51e78f0d17fc3a583ca8abc2dd118c0e |
| star_lambert.py | 4ba3d54d17e13b3267d6b010e064ef286298fa181ba23af5111c3531b93fac76 |
| star_lambert-0.1.3-source.zip | e047180d331be12291d3793bb52352e4d6bf64e81a8e660f899c5a62defc71c1 |

## 5. What it does, and its limits

Zero-revolution solver for Lambert's problem (universal variables, Stumpff functions, Newton on z; Bate-Mueller-White / Curtis Algorithm 5.2). Standard library only.

Multi-revolution transfers; transfer angles within 1e-8 rad of 0 or 180 deg (refused: the transfer plane is not defined); any statement of flight qualification.

- `lambert(r1, r2, tof, mu=398600.4418, prograde=True) → (v1, v2)`: velocities at r1 and r2 for a transfer of duration `tof`. Units: km, s, km/s, km^3/s^2.
- Never return an unconverged result: `RuntimeError` instead.
- Invalid input fails explicitly: `ValueError` for tof <= 0, zero position vectors, degenerate geometry (0 or 180 deg).

## 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_lambert_0_1_3,
  author  = {Cavazzini, Andrea},
  title   = {{star-lambert}: Universal-variable Lambert orbital boundary value problem
```

```
solver},
  version = {0.1.3},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_lambert/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts